

The Riemann Hypothesis

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1 A Function built from integers

1.1 The Riemann Zeta Function

For a complex number $s = \sigma + it$ with $\sigma > 1$, the Riemann zeta function is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1} \quad (1)$$

The sum-product identity is the Euler product and links $\zeta(s)$ with primes [1]. See the [Clay Mathematics Institute](#)¹

1.2 A classical special value

At $s = 2$, Equation 1 gives the classical identity

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots = \frac{\pi^2}{6}$$

$$\prod_{p \in \{2,3,5,7\}} \left(1 - \frac{1}{p^2}\right)^{-1} \approx 1.595 < \frac{\pi^2}{6} \approx 1.645.$$

More prime factors improve the approximation.

1.3 Two Mathematical Worlds

Figure 1 connects zeta zeros with prime distribution.

2 Zeros, Symmetry, and the Critical Line

2.1 Trivial and Nontrivial zeros

The *trivial zeros* are

$$-2, -4, -6$$

and the *non trivial zeros* lie in the *critical strip*

$$0 < \text{Re}(s) < 1$$

A Trivial Zeros

- † They occur at negative even integers.
- † Their locations are completely understood.

B Nontrivial zeros

- ⊃ They lie inside the critical strip.
- ⊃ They occur in symmetric families.
- ⊃ Their symmetries generate the following companion points:

¹The Riemann Hypothesis is a Millennium Prize Problem.

II ρ

II $1 - \rho$

II $\bar{\rho}$

II $1 - \bar{\rho}$

⊃ Their horizontal locations are central to the Riemann Hypothesis.

2.2 The functional equation

A symmetric form of the zeta function is

$$\zeta(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s), \quad \zeta(s) = \zeta(1-s) \quad (2)$$

Equation 2 reflects zeros across $\sigma = 1/2$:

$$\begin{bmatrix} \sigma' \\ t' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sigma \\ t \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The linear part of the reflection is the matrix

$$R = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad (3)$$

Hence R is an involution.

2.3 The first few zeros

The line $\text{Re}(s) = \frac{1}{2}$ is the *critical line*

Index	γ_n	real part	imaginary part
1	14.134725		14.134725
2	21.022040	$\frac{1}{2}$	21.022040
3	25.010858	$\frac{1}{2}$	25.010858
4	30.424876		30.424876

Table 1: The first four positive ordinates of non-trivial zeros.

The first two zeros form the matrix

$$Z = \begin{bmatrix} \frac{1}{2} & 14.134725 \\ \frac{1}{2} & 21.022040 \end{bmatrix} \quad (4)$$

Each row stores one zero.

2.4 A Numerical Inspection Procedure

Algorithm 1 test candidate real part

H

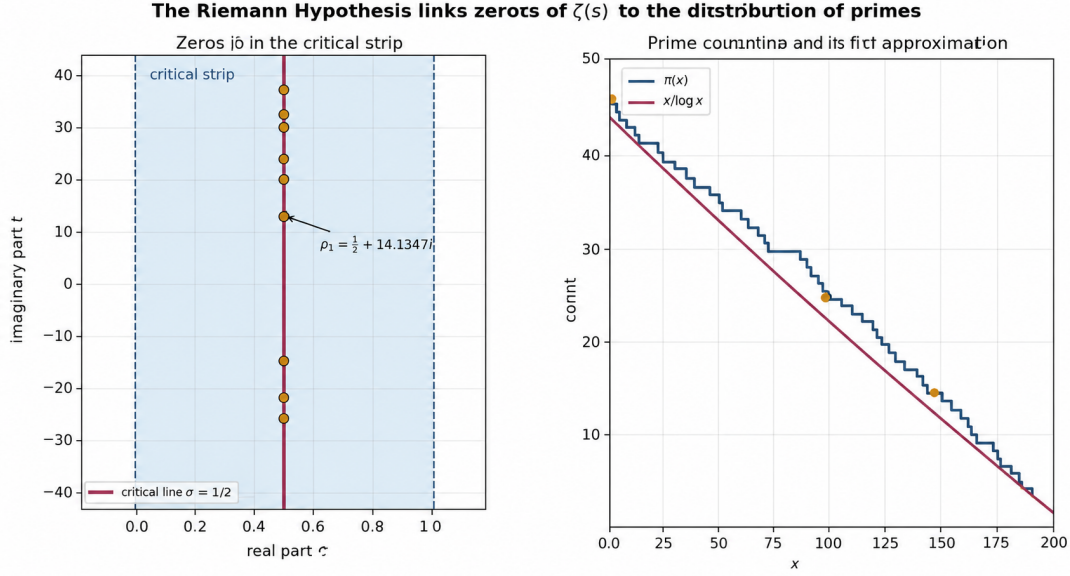


Figure 1: The critical strip and the relation between prime counting and analytic approximation.

Algorithm 1 Check candidate zeros against the critical line

Require: Candidate zeros $Z = (\rho_1, \rho_2, \dots, \rho_M)$ and tolerance $\varepsilon > 0$

Ensure: TRUE if every candidate satisfies $|\text{Re}(\rho_i) - \frac{1}{2}| \leq \varepsilon$

```

1: allOnLine ← TRUE
2: for  $i \leftarrow 1$  to  $m$  do
3:   if  $|\text{Re}(\rho_i) - \frac{1}{2}| > \varepsilon$  then
4:     allOnLine ← TRUE
5:   break
6: end if
7: end for
8: return allOnLine

```

3.3 An Equivalent Error Statement

Define the Chebyshev function

$$\psi(x) = \sum_{p^k \leq x} \log p$$

Theorem 1 (Prime-distribution error bound). *The Riemann Hypothesis is equivalent to*

$$\psi(x) = x + O(\sqrt{x} \log^2 x) \quad (6)$$

Proof. The explicit formula expresses the error through the nontrivial zeros. The stated bound is equivalent to all such zeros having real part $\frac{1}{2}$. $\square$ $\square$

Theorem 1 links zeros and prime-counting error.

3 The Riemann Hypothesis

3.1 Formal statement

The riemann ypothesis states

$$\zeta(\rho) = 0, \quad 0 < \text{Re}(\rho) < 1 \implies \text{Re}(\rho) = \frac{1}{2} \quad (5)$$

Equation 5 remains unproved[2].

3.2 Logical Interpretations

It asserts necessity, not sufficiency,

Thus,

$$\zeta(\rho) = 0 \implies \text{Re}(\rho) = \frac{1}{2}$$

for nontrivial zeros; the converse fails in general.

4 Why Prime Numbers Care

Primes appear irregularly, but zeta zeros control that irregularity. RH sharply limits the prime-counting error.

4.1 Mathematical Formulation

Let $\pi(x) = \#\{p \leq x : p \text{ is prime}\}$ denote the prime counting fnction. The prime number theorem states that

$$\pi(x) \sim \frac{x}{\log x}, \quad \lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1 \quad (7)$$

A sharper approximation is

$$\text{Li}(x) = \int_2^x \frac{dt}{\log t} \quad (8)$$

Under RH,

$$\pi(x) = \text{Li}(x) + O(\sqrt{x} \log x) \quad (9)$$

Hence zeta function govern prime counting fluctuations.

References

- [1] Harold M. Edwards. *Riemann's Zeta Function*. Academic Press, 1974.
- [2] Bernhard Riemann. "On the Number of Primes Less Than a Given Magnitude". In: *Monatsberichte der Berliner Akademie* (1859). English translation of the original German paper.